# Detection of Rare Fetal Congenital Heart Diseases (CHD) Using CARDIUM Ultrasound Images and FetalCLIP: A Deep-Learning Model for Rural Coloradans

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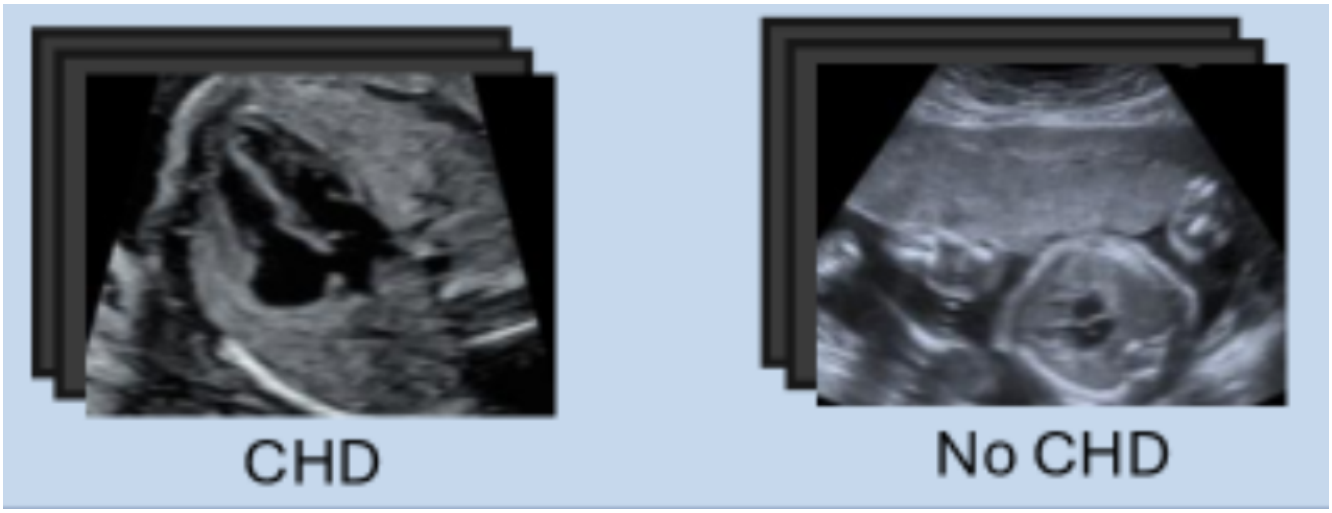



*Image source: CARDIUM Vega et al. [1]*

## Project Outline

## Abstract

Abstract

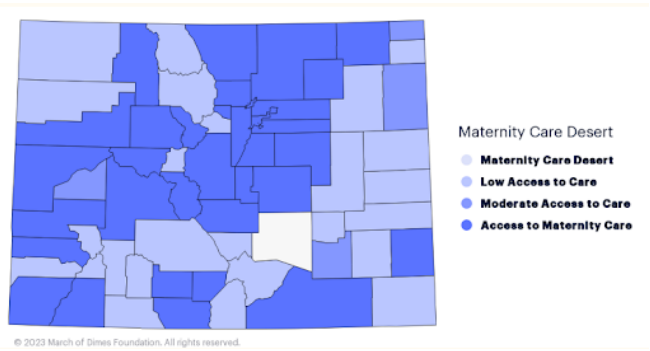
Detection of fetal congenital heart disease (CHD) is challenging in rural Colorado, particularly in areas known as Maternal Care Deserts[2], which lack access to prenatal and delivery services. In these settings, obtaining prenatal screening can be time-consuming and costly for families, motivating the need for automated risk-prediction models that support screening decisions in remote communities. We propose a multimodal deep-learning approach that combines pretrained FetalCLIP ultrasound image embeddings with CARDIUM-derived clinical feature embeddings using cross-attention fusion to estimate patient-level CHD risk. Because CHD is rare, affecting only 1% of babies, and missed cases carry significant clinical consequences, this project evaluated the impact of loss-function selection by comparing binary cross-entropy (BCE), weighted BCE, and focal loss under identical architectures and training procedures. Models were trained using three-fold cross-validation and calibrated with Platt scaling to improve probability reliability and support clinically meaningful threshold selection. Experimental results show that weighted BCE improves sensitivity but substantially increases false positives, while focal loss achieves the best overall screening trade-off and the highest precision–recall AUC. On the held-out test set, the focal loss model achieved sensitivity of 0.770 and specificity of 0.688 after calibration and threshold selection. These results demonstrate that selecting the best loss function plays an important role in improving model reliability and clinical application for rare disease detection. Focal-loss–based multimodal models provide a strong foundation for AI-assisted prenatal CHD screening in rural areas.


## 1. Introduction

Imagine you live in beautiful Rio Blanco County, Colorado and are expecting your first child. You excitedly tell your friends and family, only to find out that your county completely lacks prenatal care providers and birthing facilities. What should be a joyous time becomes a source of worry and stress. The 2024 March of Dimes report, "Nowhere to Go [2]," indicates that residents of 35% of counties throughout the United States and 40% of counties in Colorado [3] live in these "maternity care deserts," where care is unavailable. Pregnant people must choose to travel long distances across county lines to receive care, which can be expensive and dangerous in high-risk pregnancy situations or rapid deliveries.

Given that Colorado ranks 18th in the country for high-performing healthcare systems [4], one may not expect this disparity. However, significant distances between towns, often on mountain roads, and the closure of labor and delivery units force many expectant people to travel long distances for essential care. Figure 1 shows access to maternity care by Colorado county. Rural patients often experience greater medical complications as well as financial and food insecurity, contributing to higher adverse pregnancy outcomes. In "Bridging the Gap: The State of Rural Maternal Health in Colorado" [4], the Colorado Perinatal Care Quality Collaborative argues for telehealth expansion, community support programs, and improved financial reimbursement for rural hospitals. Additionally, they promote incentivizing health care professionals to practice in under-served areas, expanding training of maternal health providers, expanding postpartum care, and improving patient transportation options.

Figure 1: Colorado Maternity Care Deserts




Patients living in maternity care deserts balance their baby's health with their own health, time, finances, and care for their other children. Understanding their risks prior to delivery allows patients to coordinate the level of prenatal and neonatal care they need. For example, if a patient learns that their baby may have a CHD, they may choose to temporarily relocate to Denver, where Level IV neonatal intensive care units are available. However, receiving the initial screening can be expensive and chal-

lenging, particularly because access to trained fetal cardiac imaging specialists is limited or non-existent in rural areas. As such, there is an opportunity to use automated image-based risk prediction models to support earlier screening decisions closer to where patients live.

In the future, we hope to create a tool for traveling sonographers to be used in Colorado's maternity care deserts. If the risk score exceeds a certain threshold, the patient would be referred for additional follow-up. For this class project, our scope was limited to designing and evaluating a deep-learning model used to generate the CHD risk score. We are not building the full screening application, but we describe the intended deployment setting, as it was used to guide model design decisions such as minimizing missed CHD cases.

## 2. Related Work

### 2.1 Topic 1: Artificial Intelligence (AI) for Prenatal Ultrasound Detection of CHD

Several studies have evaluated whether AI can improve the identification of prenatal CHD when compared with fetal-heart experts [1] [5] [6] [7] [8] [9]. They found that AI improves overall accuracy and provides a powerful supportive tool that augments clinical diagnosis. CARDIUM obtained a CHD F1 Score of $0.798 \pm 0.048$, precision of $0.876 \pm 0.173$, recall of $0.757 \pm 0.104$ and AUC of $0.974 \pm 0.012$. FetalCLIP achieved an AUC of 74.66 for still ultrasound images of CHD. Like our study, the authors hope to see AI expand access for patients in remote areas. Our work differs in that we focused specifically on reducing false negatives in prenatal CHD screening under class imbalance by evaluating loss-function choices such as weighted loss and focal loss, which directly target improved sensitivity for clinically important cases that might otherwise be missed. Because missed prenatal CHD diagnoses can delay referral, delivery planning, and access to specialized neonatal care, improving sensitivity was a primary objective of our work.

### 2.2 Topic 2: AI-Assisted Clinical Decision Support and Deployment in CHD Screening

Prior studies describe the integration of AI tools into clinical workflows to support screening, improve accuracy, and assist with prenatal decision-making [7] [10] [11]. These papers also leverage AI in identifying high-risk patients who may require tailored interventions to reduce mortality and morbidity. BRIGHT-HEART provides an FDA-approved AI tool for identifying CHD that hospitals can purchase. Unlike these studies, our project emphasized improving access to care by enabling initial screening where patients live and by prioritizing sensitivity to reduce missed CHD cases before a case is referred for specialist review.

### 2.3 Topic 3: Data Limitations and Multimodal Learning in AI-Based CHD Detection

Related studies have identified limited training data, dataset heterogeneity, and challenges with model generalization as major barriers to reliable clinical deployment of AI systems for CHD detection [8] [12] [1] [5]. As with our study, the papers explored multimodal learning frameworks that integrate ultrasound imaging with maternal clinical data and CLIP-based image representations. Our study differs in that we focus on explicitly evaluating whether class-imbalance-aware loss functions can improve sensitivity and reduce false negatives in prenatal CHD risk prediction.

## 3. Methods

### 3.1 Problem Formulation

We aimed to develop a deep-learning model that estimated prenatal risk of CHD using ultrasound and echocardiography imaging and maternal clinical data. In the long term, this model could serve as the core component of a prenatal decision-support platform designed for deployment in maternity care deserts, where high-risk scores would trigger specialist review and additional follow-up, as needed. For this project, our focus was on designing and evaluating the CHD risk prediction model itself rather than building the full deployment platform. The intended clinical use context motivates our emphasis

on producing risk predictions and interpretable supporting clinical factors (such as advanced maternal age), rather than a simple Yes/No response.

### 3.2 Data / Inputs

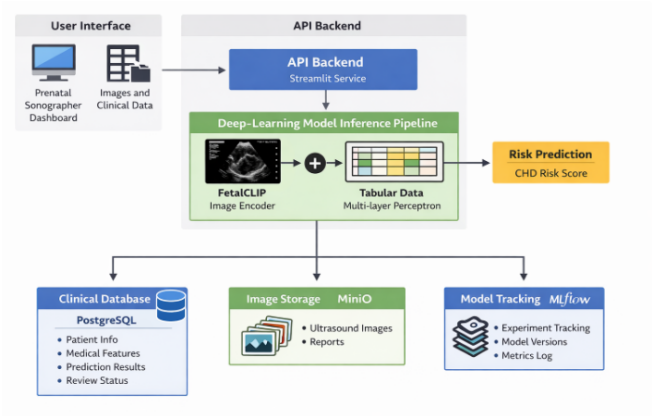
This project employed the publicly available and de-identified CARDIUM dataset of ultrasound images and maternal clinical data for CHD detection [1]. CARDIUM oversampled CHD babies with a prevalence of 7.19%, compared to 1% in the general population. Data were collected through a retrospective study from 2013 to 2024 of 1,103 Colombian patients with either obstetric echocardiographic or ultrasound images and associated clinical records available. Patients were over 18 years old and had singleton pregnancies. In total, CARDIUM included 6,558 images with 16.3% belonging to CHD patients and 83.7% corresponding to patients without CHD. The prediction target is patient-level CHD risk rather than image-level classification. Multiple images for the same patient are linked through an ID table.

CARDIUM provided preprocessed data which included normalized tabular data and organized in three folds. Each fold included all of the data with random splits for training, validation and test sets. CARDIUM performed all data splits at the patient level to prevent data leakage, ensuring that images from the same patient did not appear across training, validation, and test sets within the same fold. Model predictions aggregated results at the patient level using mean pooling to align with clinical decision-making.

### 3.3 System and Model Architecture

The proposed future tool that will be used by prenatal sonographers will take inputs of ultrasound images and brief patient history and process these data through a backend deep-learning pipeline (Figure 2). The model combines image understanding (FetalCLIP) with clinical features and outputs a CHD risk score. In a future deployment setting, if the patient provides consent for data storage, the system could store structured data, images, and model outputs in a clinical database and support monitoring of model performance for continuous improvement. The scope of this class project included the green box for "Deep-Learning Model Inference Pipeline" and the yellow box for "Risk Prediction."

Figure 2: System Architecture



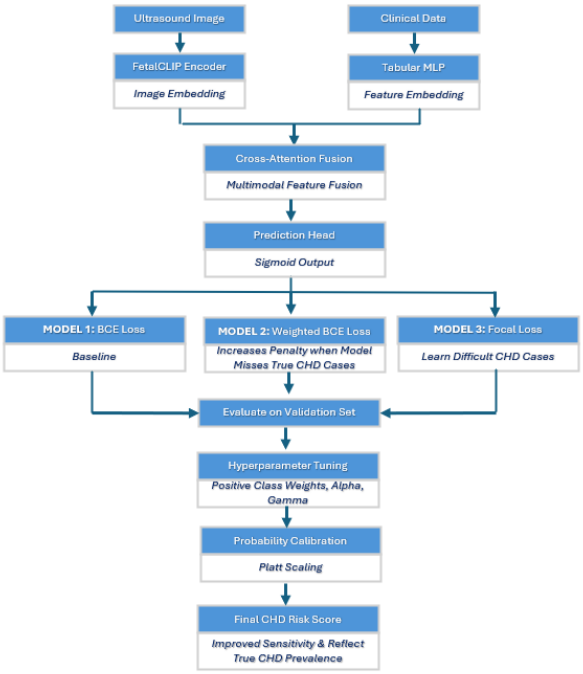

As illustrated in Figure 3, the model used in the deep-learning pipeline extracted image features using FetalCLIP (which has pretrained weights), processed clinical/tabular data using the CARDIUM approach, combined both using cross-attention fusion, evaluated the performance of three different loss functions, calibrated the probabilities, completed threshold tuning, and predicted CHD risk. Cross-attention fusion layers allowed the model to learn interactions between ultrasound image embeddings and maternal clinical tabular embeddings before final risk prediction.

The FetalCLIP encoder was frozen and used as a fixed feature extractor. We used the image encoder portion of FetalCLIP to extract 767 fixed-length feature embeddings for each image. Image embeddings were computed once and saved so they could be reused across experiments, which improved computational efficiency during model development. FetalCLIP helped the model understand patterns in the fetal heart.

For the tabular data, the CARDIUM encoder was frozen and used as a fixed feature extractor. We used the tabular encoder portion of the existing CARDIUM approach to extract 127 fixed-length feature embeddings from the clinical data. Text embeddings were computed once and saved so that they could be reused across experiments. As with the image embedding, these text encodings improved computational efficiency during model development.

We anticipated that using weighted BCE and focal loss to penalize missed CHD cases and learn difficult CHD cases would help the models identify

Figure 3: Model Architecture




false negatives and learn from them. We adjusted hyperparameters, as needed, including prediction threshold to increase sensitivity (recall). The model probability outputs were calibrated using Platt scaling to ensure that predicted risk scores reflect the true prevalence of coronary heart disease and support the selection of clinically meaningful referral thresholds [13]. As we complete the modeling exercises, we assessed how well Platt scaling worked with our models by comparing the distributions of the raw and calibrated predictions, generating the class-wise distributions for CHD and non-CHD cases, and calculating the Brier score for the raw and calibrated data. For binary outcomes, the Brier score measures how close predicted probabilities are to the true labels by computing the mean squared difference between each predicted probability and the actual outcome (0 or 1) [14]. Lower values indicate better calibration and overall probabilistic accuracy. Brier score evaluates the quality of the probability estimates themselves, making it especially useful for assessing whether a model's predicted risks are trustworthy for decision-making. If the Brier score for the calibrated data was lower than the Brier score for the raw data, that change indicated the the calibration had helped to make the predictions more reliable.

## 4. Experiments

Our biggest concern was that our model would miss positive CHD cases. Missed cases who are delivered at a low-level facility risk the baby's survival, as they will not have immediate access to high-level care and instead will require transfer. As such, we focused our experiments on two loss functions, which may help to improve detection of rare CHD cases. Specifically, we compared a baseline binary cross-entropy (BCE) loss model with weighted BCE loss and focal loss.

The loss functions we used have been well-studied and can be summarized as follows [15]:

- *Binary Cross-Entropy (BCE)* measures the difference between predicted probabilities and true binary labels by penalizing incorrect predictions more strongly when the model is confident but wrong. BCE treats all samples equally, which can lead to poor performance when classes are highly imbalanced because BCE weights mistakes for either class equally.
- *Weighted Binary Cross-Entropy (Weighted BCE)* modifies BCE by assigning larger penalties to missed positive cases during training by increasing the weights for this group and down-weighting the easy negative cases. This approach encourages the model to pay more attention to underrepresented outcomes such as rare disease cases.
- *Focal Loss* extends BCE by down-weighting easy-to-classify examples and focusing learning on harder or misclassified samples. This approach is especially helpful in imbalanced datasets. Focal loss improves robustness by reducing false negatives in rare disease cases and preventing the majority class from dominating the training signal.

**Evaluation metrics:** The primary evaluation metric was sensitivity (recall) for CHD cases, since we focused on minimizing false negatives. Secondary evaluation metrics included precision (to monitor false positive burden), area under the ROC curve (AUC), precision-recall curve (PR-AUC) and calibration performance after Platt scaling and threshold tuning.

Data were split at the patient level into training (70%), validation (15%), and test (15%) sets. All models used identical architectures, pretrained Fe-

talCLIP image embeddings, clinical feature inputs, and optimization procedures. Specifically, in order to isolate the effects of the loss functions, the model parameters and evaluation metrics for the three approaches were held constant. Models were trained using the Adam optimizer with early stopping based on validation loss. Hyperparameters, including class weights and prediction thresholds, were selected using validation-set performance to prioritize recall and reduce false negatives. Final evaluation was performed on the held-out test set. Performance differences between models were assessed by comparing sensitivity and PR-AUC across experimental conditions.

### 4.1 Experiment 1: Weighted Binary Cross-Entropy Loss

**Main purpose:** The goal of this experiment was to evaluate whether weighted BCE improves sensitivity for CHD detection compared with BCE loss and focal loss.

**Experimental setup:** We trained a baseline multimodal model using BCE loss and compared it with a model trained using weighted BCE loss. The positive-class weight was initially set proportional to the inverse class prevalence for the images and then refined using validation-set performance. For hyperparameter tuning, we tested positive weights of 5, 10, 14, 20.

### 4.2 Experiment 2: Focal Loss for Hard Example Learning

**Main purpose:** The goal of this experiment was to evaluate whether focal loss improved detection of difficult-to-classify CHD cases compared with BCE loss.

**Experimental setup:** The baseline multimodal model trained with BCE loss and the model trained with weighted BCE loss were compared with a model trained using focal loss. The focal loss parameters $\alpha$ (class weighting) and $\gamma$ (focusing strength) were initially set at $\alpha$=0.25 and $\gamma$=2 and then tuned using validation-set performance. For hyperparameter tuning, we tested the following:

- $\alpha = 0.25, \gamma = 1$
- $\alpha = 0.25, \gamma = 2$
- $\alpha = 0.50, \gamma = 1$
- $\alpha = 0.50, \gamma = 2$
- $\alpha = 0.75, \gamma = 2$
- $\alpha = 0.75, \gamma = 3$

### 4.3 Experiment Results

CARDIUM provides three predefined folds with randomized splits for training, validation, and testing. Each fold contains the full dataset of 6,558 images for 1,103 individual patients. Three-fold cross-validation was performed using these splits, and the average performance across folds is reported in Table 1. This table summarizes model performance for the evaluated loss functions and corresponding hyperparameter settings. All models used the same underlying architecture and training configuration unless otherwise specified.

For each loss function family, the best configuration was selected based on precision–recall area under the curve (PR-AUC). The baseline BCE model achieved high accuracy (0.945) and moderate precision (0.706), but low sensitivity (recall) (0.408), indicating that many positive CHD cases were missed. Increasing the positive class weight in weighted BCE improved sensitivity, with the best configuration (pw = 10) achieving sensitivity of 0.757 at the expense of reduced precision (0.252) and accuracy (0.738). Among the focal loss configurations, the model with $\alpha = 0.75$ and $\gamma = 2$ achieved the highest PR-AUC and resulted in a more balanced model. Sensitivity (0.608) and precision (0.551) were moderate, while accuracy was high (0.937).

The model probability outputs were calibrated using Platt scaling to improve agreement between predicted risk scores and the observed outcome frequencies, supporting the selection of clinically meaningful referral thresholds [13]. Figure 4 illustrates the probability distributions of the raw and calibrated predictions for the best-performing configuration of each loss function. Across all models, Platt scaling produced a narrower probability range and shifted predictions toward lower values relative to the raw outputs, indicating improved probability calibration and reduced overconfidence in high-risk predictions. These results are supported by the Brier scores, which decreased for the calibrated predictions compared to the raw predictions for both weighted BCE and focal loss (Table 2).

Table 1: Cross-validation performance comparison of candidate loss functions prior to probability calibration.

| Model | AUROC | PR-AUC | Sensitivity | Precision | F1 | Accuracy |
|---|---|---|---|---|---|---|
| BCE | 0.787 ± 0.026 | 0.523 ± 0.012 | 0.408 ± 0.221 | 0.706 ± 0.163 | 0.472 ± 0.180 | 0.945 ± 0.010 |
| Weighted BCE (pw=5) | 0.829 ± 0.016 | 0.498 ± 0.033 | 0.447 ± 0.254 | 0.584 ± 0.198 | 0.436 ± 0.148 | 0.931 ± 0.016 |
| Weighted BCE (pw=10) | 0.843 ± 0.017 | 0.573 ± 0.007 | 0.757 ± 0.104 | 0.252 ± 0.201 | 0.339 ± 0.184 | 0.738 ± 0.166 |
| Weighted BCE (pw=14) | 0.832 ± 0.006 | 0.529 ± 0.008 | 0.594 ± 0.041 | 0.580 ± 0.158 | 0.576 ± 0.065 | 0.939 ± 0.021 |
| Weighted BCE (pw=20) | 0.835 ± 0.030 | 0.529 ± 0.032 | 0.660 ± 0.171 | 0.396 ± 0.206 | 0.448 ± 0.125 | 0.866 ± 0.109 |
| Focal ($\alpha = 0.25, \gamma = 1$) | 0.837 ± 0.025 | 0.552 ± 0.030 | 0.406 ± 0.200 | 0.812 ± 0.037 | 0.520 ± 0.188 | 0.954 ± 0.011 |
| Focal ($\alpha = 0.25, \gamma = 2$) | 0.831 ± 0.010 | 0.546 ± 0.030 | 0.514 ± 0.134 | 0.681 ± 0.322 | 0.539 ± 0.148 | 0.936 ± 0.041 |
| Focal ($\alpha = 0.50, \gamma = 1$) | 0.837 ± 0.015 | 0.524 ± 0.038 | 0.353 ± 0.272 | 0.628 ± 0.164 | 0.414 ± 0.299 | 0.945 ± 0.014 |
| Focal ($\alpha = 0.50, \gamma = 2$) | 0.834 ± 0.024 | 0.561 ± 0.043 | 0.567 ± 0.030 | 0.639 ± 0.170 | 0.596 ± 0.088 | 0.947 ± 0.015 |
| Focal ($\alpha = 0.75, \gamma = 2$) | 0.830 ± 0.024 | 0.607 ± 0.047 | 0.608 ± 0.029 | 0.551 ± 0.170 | 0.571 ± 0.107 | 0.937 ± 0.023 |
| Focal ($\alpha = 0.75, \gamma = 3$) | 0.833 ± 0.022 | 0.508 ± 0.086 | 0.364 ± 0.248 | 0.638 ± 0.156 | 0.408 ± 0.143 | 0.937 ± 0.011 |

Table 2: Validation-set Brier scores before and after Platt scaling calibration.

| Model | Raw Brier Score | Calibrated Brier Score |
|---|---|---|
| BCE | 0.0406 | 0.0459 |
| Weighted BCE | 0.2369 | 0.0594 |
| Focal Loss | 0.0709 | 0.0484 |

Lower values indicate better probabilistic calibration.

Once Platt scaling was applied to the validation-set predictions, revised decision thresholds for CHD detection were selected to favor high sensitivity for the BCE and focal loss models. For weighted BCE, the threshold corresponds to the calibrated equivalent of a raw probability threshold of 0.5. The resulting performance is summarized in Table 3.

The BCE model achieved the highest sensitivity (0.909), corresponding to a false negative rate of 0.091, but this occurred at the expense of reduced specificity (0.457). The focal loss model achieved the next-highest sensitivity (0.818) with improved specificity (0.617) compared with BCE, indicating a more balanced trade-off between missed CHD cases and false positives. In contrast, weighted BCE achieved lower sensitivity (0.636) despite higher specificity (0.691), resulting in a relatively high false negative rate (0.364), which may limit its suitability for screening applications prioritizing case detection.

Next, model performance was evaluated on the held-out test set. As shown in Table 4, the focal loss model provided the most balanced performance, achieving a sensitivity of 0.770 and specificity of 0.688. Confusion matrices for the models are provided in Figure 5.

In total, the BCE model missed 15 CHD cases and produced 524 false positives. Although weighted BCE reduced the number of missed CHD cases to 8 false negatives, it resulted in nearly 600 false positives, substantially increasing the burden of additional referrals and potential anxiety for families. In comparison, the focal loss model missed only two additional CHD cases relative to BCE (17 false negatives) while reducing the number of false positives to 321 patients. Although still high, this represents a more favorable trade-off between case detection and unnecessary follow-up testing in a screening context.

Differences in precision–recall performance across models are illustrated in Figure 6, where the focal loss model achieved the highest AUPRC at 0.522.

### 4.4 General Trends

Across experiments, focal loss performed the best. It resulted in the most balanced screening performance after calibration and threshold selection, achieving strong sensitivity and reducing false positives relative to BCE and weighted BCE. Although weighted BCE achieved the highest sensitivity on the test set, this improvement came at the cost of a large increase in false positive predictions. Overall, focal loss achieved the best trade-off between sensitivity and specificity while also producing the highest AUPRC among the evaluated models.

In the CARDIUM dataset, only 7.19% of patients had CHD. Our results suggest that selecting the best

Table 3: Validation-set operating-point performance after Platt calibration.

| Loss Function | Threshold[a] | Sensitivity | Specificity | Precision | F1 Score | FNR |
|---|---|---|---|---|---|---|
| BCE | 0.050 | 0.909 | 0.457 | 0.106 | 0.190 | 0.091 |
| Focal Loss | 0.050 | 0.818 | 0.617 | 0.131 | 0.226 | 0.182 |
| Weighted BCE | 0.071 | 0.636 | 0.691 | 0.127 | 0.212 | 0.364 |

[a] Threshold values correspond to calibrated probability thresholds after applying Platt scaling to validation-set predictions. For weighted BCE, the threshold represents the calibrated equivalent of a raw probability threshold of 0.5.
BCE = Binary cross-entropy loss; FNR = false negative rate.

Table 4: Test-set operating-point performance after Platt calibration.

| Loss Function | Threshold[a] | AUROC | AUPRC | Sensitivity | Specificity | Precision | F1 | FNR |
|---|---|---|---|---|---|---|---|---|
| BCE | 0.050 | 0.773 | 0.480 | 0.797 | 0.491 | 0.101 | 0.180 | 0.203 |
| Weighted BCE | 0.071 | 0.799 | 0.327 | 0.757 | 0.737 | 0.171 | 0.279 | 0.243 |
| Focal Loss | 0.050 | 0.814 | 0.522 | 0.770 | 0.688 | 0.151 | 0.252 | 0.230 |

[a] Threshold values correspond to calibrated probability thresholds after applying Platt scaling to model predictions using the validation set. For weighted BCE, the threshold represents the calibrated equivalent of a raw probability threshold of 0.5.
BCE = Binary cross-entropy loss; AUROC = area under the receiver operating characteristic curve; AUPRC = area under the precision–recall curve; FNR = false negative rate.

loss-function can result in improved sensitivity and specificity in imbalanced medical classification tasks such as CHD detection.

### 4.5 Insights into These Results

Several factors help explain why focal loss performed better than BCE and weighted BCE in this setting. First, the dataset is strongly class imbalanced. Unweighted BCE treats all samples equally during optimization and therefore tends to favor the majority class, resulting in lower sensitivity. Weighted BCE partially addresses this by increasing the contribution of positive examples to the loss function; however, large positive weights can cause the model to overcompensate and produce excessive false positives. Focal loss improves performance in this setting because it down-weights easy negative examples while focusing training on harder to classify samples. This allows the model to better distinguish between non-CHD and CHD cases without excessively increasing false positive predictions. As a result, focal loss achieved a more favorable balance between missed detections and unnecessary follow-up screening.

Additionally, probability calibration using Platt scaling improved the distribution of predicted risk scores and enabled consistent selection of clinically meaningful operating thresholds across models. Calibration also reduced overconfident probability estimates closer to zero and one.

### 4.6 Remaining Questions

Although hyperparameters were tuned for weighted BCE and focual loss, it is possible that we would have achieved a better model with additional tuning. For example, focal loss with $\alpha = 0.75$ and $\gamma = 2$ performed the best. However, additional combinations of $\alpha$ and $\gamma$ may further improve performance or produce different sensitivity–specificity trade-offs. Similarly, alternative calibration approaches such as isotonic regression were not explored and may provide improved probability reliability. Additionally, model evaluation was performed using cross-validation splits derived from CARDIUM, as this dataset is publicly available. A remaining question is how this model would perform on external datasets to confirm generalizability across populations, imaging protocols, and clinical environments.

Of note, we were unable to identify publicly available datasets containing fetal cardiac ultrasound images from Colorado residents. Because geographic

Figure 4: Validation Probability Distributions for Raw and Calibrated Predictions by Loss Function

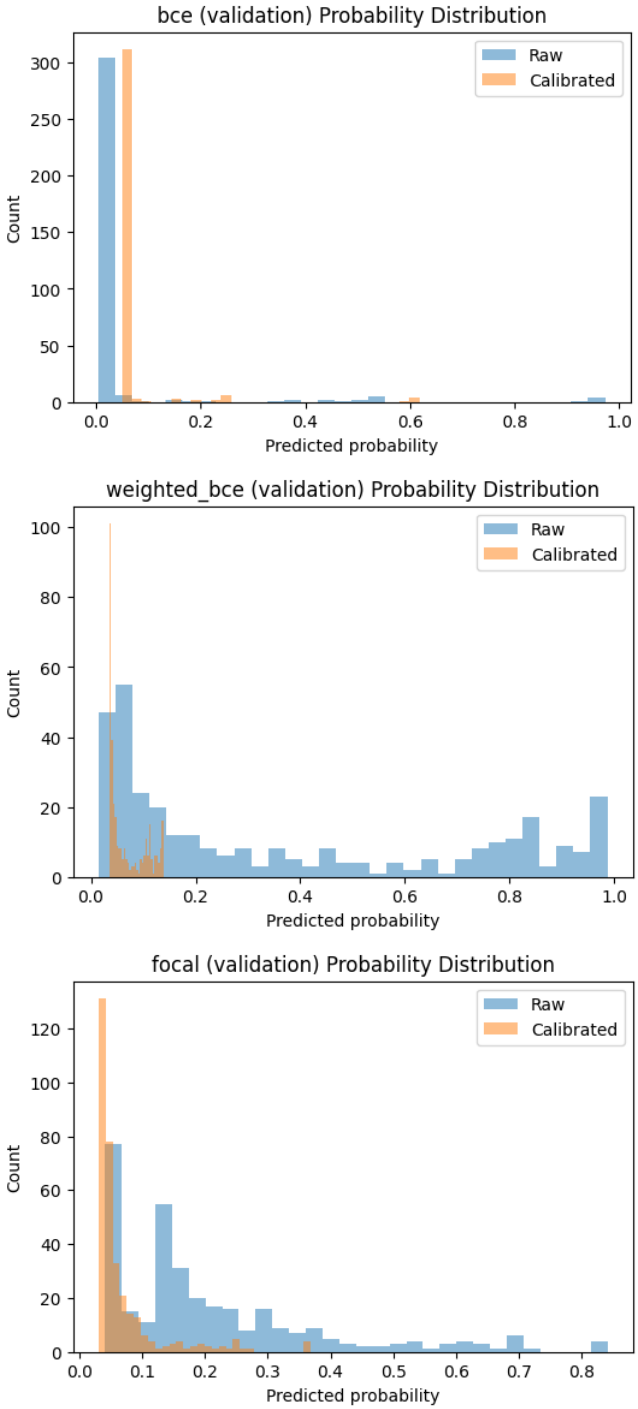



Figure 5: Test-set Confusion Matrices for Final Models with Platt Scaling

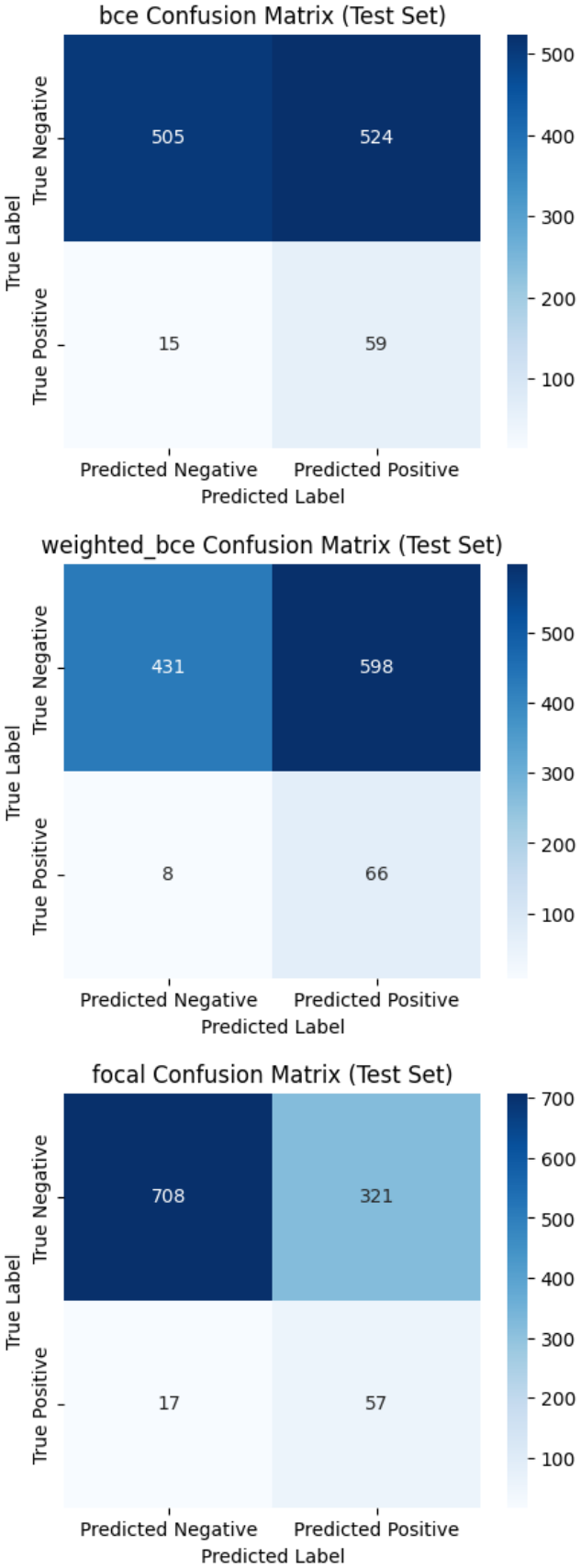
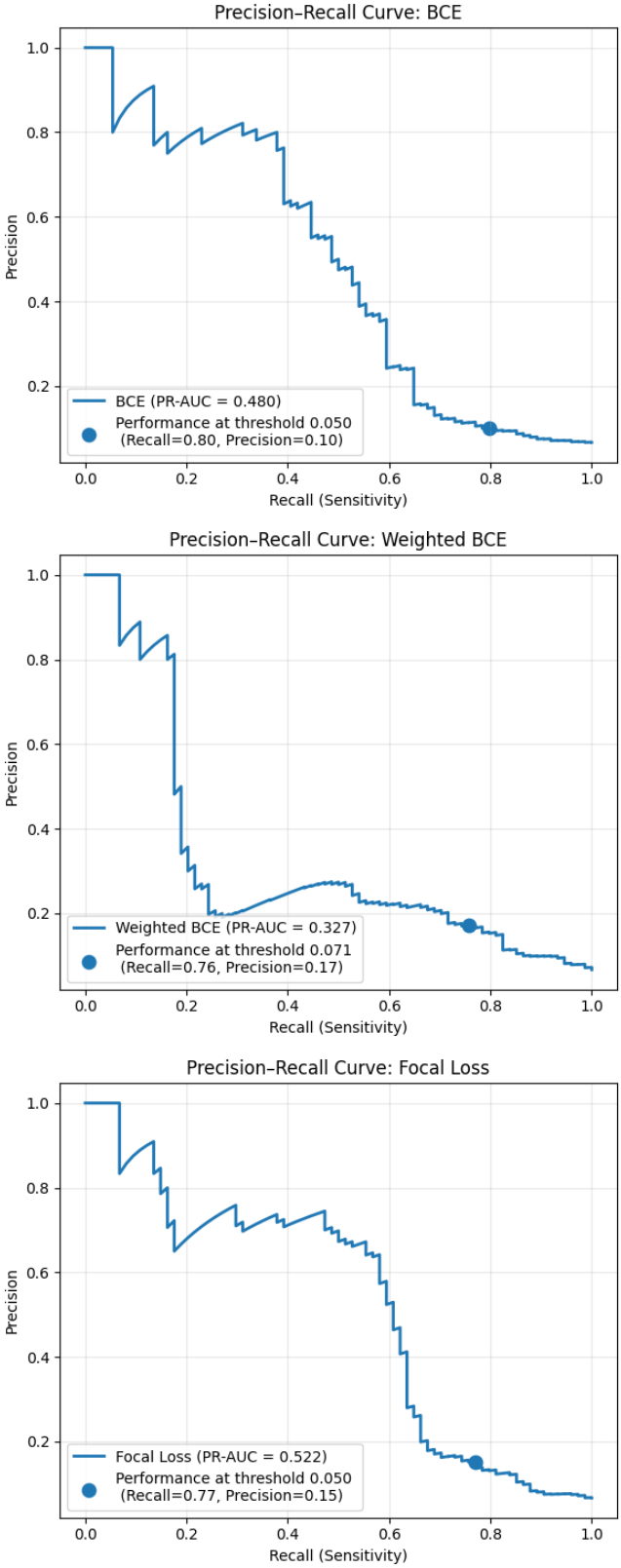


Figure 6: Test-set Precision-Recall Curves for Final Models with Platt Scaling




and population differences may affect model generalizability, future work could involve collaboration with local health systems to evaluate model performance on data from Colorado populations if this screening approach proves promising. Additionally, we had planned to subset to patients to those with ultrasound images because at approximately $260 [16] ultrasound is relatively inexpensive compared to echocardiogram, and if the model is used for the traveling prenatal sonographer tool, the provider may not have prenatal echocardiogram training [17]. However, the CARDIUM dataset did not include an indicator for the type of image (i.e. ultrasound or echocardiogram). As such, we have included both image types in this analysis.

Finally, further investigation of the missed cases is required to better understand the severity and mortality risks for these cases. Due to the complexity of CHD, we anticipate that working with a fetal heart specialist for this review would be most useful.

### 4.7 Next Steps

Several extensions of this work could be done to strengthen the application of this model. Ideally, a model exists that captures almost all of the CHD cases, without a heavy burden of false positives for additional follow-up. A promising direction may be to explore alternative probability calibration techniques, such as isotonic regression or temperature scaling, and compare their impact on screening performance. Future work could also investigate optimizing the decision threshold directly for clinically defined operating targets, such as maintaining sensitivity above a fixed screening requirement while minimizing false positives. This would further align model behavior with real-world deployment requirements.

As mentioned in the remaining questions, one important next step would be to evaluate the selected focal loss model on an independent external dataset to assess robustness and generalizability. External validation is particularly important in medical imaging applications where performance can vary across acquisition settings and patient populations.

## 5. Conclusions

This study aimed to create a multimodal deep learning model for fetal CHD detection model that would capture as many CHD cases as possible without over burdening sonographers with high numbers of false positives. The project evaluated the impact of loss-function selection, specifically BCE, weighted BCE and focal loss to reduce false negatives. Across experiments, focal loss produced the best model after Platt calibration and threshold selection, achieving strong sensitivity while substantially reducing false positives relative to BCE and weighted BCE. The focal loss model achieved a sensitivity (recall) of 0.770, specificity of 0.688, and AUC of 0.522. When compared to CARDIUM, which achieved a recall of 0.757, our model obtained a slightly higher recall score. Our AUC score was lower than both CARDIUM (0.974) and FetalCLIP (0.747).

These results demonstrate that selecting the best loss function plays an important role in improving model reliability and clinical application for rare diseases. Because this model is intended for screening a vulnerable population with prenatal screening, several ethical aspects should be considered. False negative predictions may delay diagnosis and treatment for newborns with CHD, while false positives may increase unnecessary referrals, cost and anxiety for families. Ensuring equitable performance across patient populations is also critical, particularly given known disparities in access to specialized prenatal imaging in maternity care deserts. Future work should include evaluation of subgroup performance across demographic and geographic populations, as well as external validation on independent datasets. Finally, training for sonographers, clinicians and staff should focus on how to best communicate the model outputs to patients and how these outputs influence decision-making. With appropriate validation and safeguards, models such as this may help improve early CHD detection in settings where expert fetal cardiology resources are absent or limited.

## Bibliography

## References


[1] D. Vega, H. V. Ceballos, J. S. Vera, S. Rodriguez, A. Perez, A. Castillo, M. Escobar, D. Londoño, L. A. Sarmiento, C. I. Castro, N. Rodriguez, J. C. Briceño, and P. Arbeláez, "Cardium: Congenital anomaly recognition with diagnostic images and unified medical records," *arXiv preprint arXiv:2510.15208*, 2025.

[2] March of Dimes, "Nowhere to go: Maternity care deserts across the us 2024 report," 2024.

[3] R. Driscoll, "Maternal health care deserts in colorado," November 2022.

[4] Colorado Perinatal Care Quality Collaborative (CPCQC), "Bridging the gap: The state of rural maternal health in colorado," 2024.

[5] F. Maani, N. Saeed, T. J. Saleem, Z. Farooq, H. Alasmawi, W. Diehl, A. Mohammad, G. Waring, S. Valappil, L. Bricker, and M. Yaqub, "Fetalclip: A visual-language foundation model for fetal ultrasound image analysis," *arXiv preprint arXiv:2502.14807*, 2025.

[6] S. Jassim and S. Basheer, "Ai vs. traditional ultrasound study in congenital heart defect detection: A systematic review," *medRxiv*, 2025.

[7] S. G. Niyogi, D. S. Nag, M. M. Shah, A. Swain, C. Naskar, P. Srivastava, and R. Kant, "Role of artificial intelligence in congenital heart disease," *World Journal of Clinical Pediatrics*, vol. 14, no. 3, p. 105926, 2025.

[8] K. Kaur, C. Singh, and Y. Kumar, "Diagnosis and detection of congenital diseases in newborns or fetuses using artificial intelligence techniques: A systematic review," *Archives of Computational Methods in Engineering*, vol. 30, pp. 3031–3058, 2023.

[9] K. Khan, F. Ullah, I. Syed, and H. Ali, "Accurately assessing congenital heart disease using artificial intelligence," *PeerJ Computer Science*, vol. 10, p. e2535, 2024.

[10] BrightHeart, "Brightheart secures fda clearance for first ai software revolutionizing prenatal fetal heart ultrasound evaluations," 2024.

[11] Mount Sinai Health System, "Mount sinai obstetricians first in new york city to use fda-cleared ai tool that enhances fetal screenings for congenital heart defects," 2025.

[12] H. Ejaz, T. Thyyib, A. Ibrahim, A. Nishat,

and J. Malay, "Role of artificial intelligence in early detection of congenital heart diseases in neonates," *Frontiers in Digital Health*, vol. 5, p. 1345814, 2024.

[13] Train in Data, "The complete guide to platt scaling." Train in Data Blog, 2025. Accessed April 1, 2026.

[14] J. Goldstein-Greenwood, "A brief on brier scores." University of Virginia Library StatLab, 2021. Accessed April 17, 2026.

[15] O. Elharrouss, Y. Mahmood, Y. Bechqito, M. A. Serhani, E. Badidi, J. Riffi, and H. Tairi, "Loss functions in deep learning: A comprehensive review," 2025. arXiv preprint.

[16] Sidecar Health, "Cost of prenatal ultrasound by state," 2024. Accessed: 2026-03-29.

[17] American Heart Association, "Fetal echocardiogram test," 2023. Accessed: 2026-03-29.

[18] A. Pozza, L. Zanella, B. Castaldi, and G. Di Salvo, "How will artificial intelligence shape the future of decision-making in congenital heart disease?," *Journal of Clinical Medicine*, vol. 13, no. 10, p. 2996, 2024.

[19] I. R. de Vries, J. O. E. H. van Laar, M. B. van der Hout-van der Jagt, S.-A. Clur, and R. Vullings, "Fetal electrocardiography and artificial intelligence for prenatal detection of congenital heart disease," *Acta Obstetricia et Gynecologica Scandinavica*, vol. 102, pp. 1511–1520, 2023.

[20] H. Tantai, "Use weighted loss function to solve imbalanced data classification problems." Medium, 2023. Blog post accessed April 1, 2026.

[21] J. C. Platt, "Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods," in *Advances in Large Margin Classifiers* (A. J. Smola, P. L. Bartlett, B. Schölkopf, and D. Schuurmans, eds.), pp. 61–74, MIT Press, 1999.